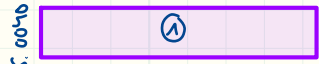
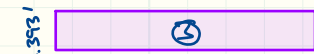


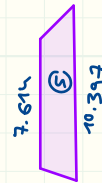
Directional StabilityS_{D,S} Calculation:

$58.4292'$

 $A_1 = 350.7969 \text{ ft}^2$

$37.3707'$

 $A_2 = \frac{1}{2} (37.3)(3.51) = 65.615 \text{ ft}^2$

$4.393'$

 $A_3 = (4.393)(21.0865) = 92.633 \text{ ft}^2$


 $A_4 = \frac{(10.399 + 7.614) \times 2.1965}{2}$
 $= 19.7806 \text{ ft}^2$


 Approx as half circle
 $r = 3.807 \text{ ft}$

$A_5 = \frac{\pi (3.807)^2}{2} = 22.766 \text{ ft}^2$

$S_{D,S} = A_1 + A_2 + A_3 + A_4 + A_5 = 551.5923 \text{ ft}^2$

Aircraft Data:

$l_f = 65.0167 \text{ ft}$

$b_w = 93.84 \text{ ft}$

$S_{D,S} = 551.5923 \text{ ft}^2$

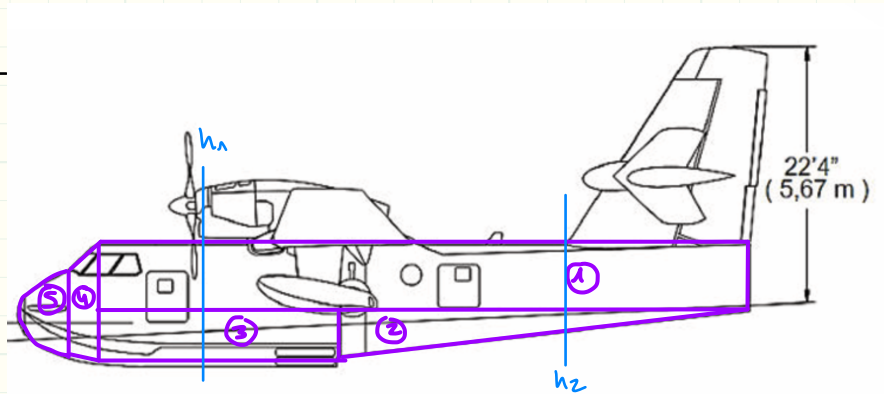
$\Gamma = 0^\circ$

$\alpha = 0^\circ$

$S_w = 1023.95$

$AR_w = 8.6$

$\lambda = 1$



Fuselage Contribution

$$(C_{n\beta})_f = -K_N K_{RL} \left(\frac{S_{B,S}}{S_w} \right) \left(\frac{l_f}{b} \right)$$

K_N :

$$\frac{x_m}{l_f} = 0.39$$

$$\frac{l_f^2}{S_{B,S}} = 7.66$$

$$\sqrt{\frac{h_1}{h_2}} = 1.18$$

$$\frac{h}{b_{f,max}} = 1.21$$

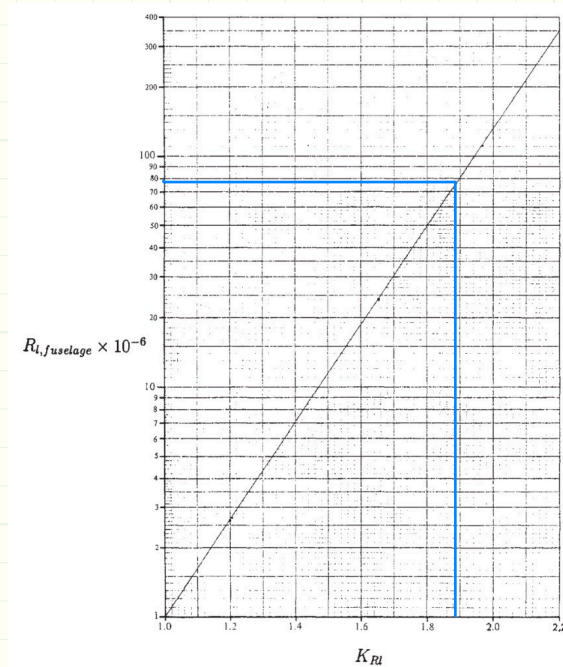


Fig. 3.74 Variation of K_{RL} with fuselage Reynolds number.¹

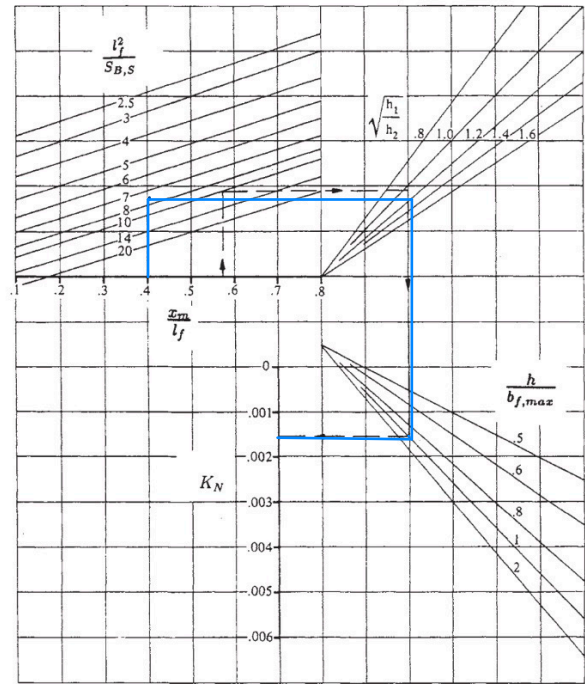
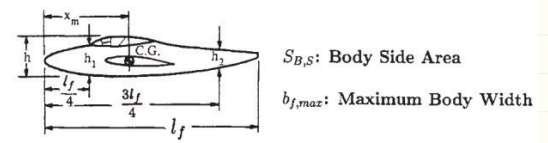


Fig. 3.73 Empirical factor K_N related to $(C_{n3})_{B(W)}$ (Ref. 1).

$$Re_f = \frac{V l_f}{\nu} = 7.85 \times 10^7$$

$$(C_{n\beta})_f = -0.0015 (1.89) \left(\frac{551.6}{1023.95} \right) \left(\frac{55.0167}{93.84} \right) = -0.0013 \text{ /deg}$$

Dihedral & Sweep Contribution are zero since

$$\Gamma = 0 \quad \& \quad \Lambda = 0$$

• V.T Contribution

$$(C_{n_p})_v = K C_{L\alpha_v} \left(1 + \frac{\partial \sigma}{\partial \beta}\right) \eta_v \left(\frac{S_v}{S}\right) \left(\frac{b_v}{b_w}\right)$$

$$\left(1 + \frac{\partial \sigma}{\partial \beta}\right) \eta_v = 0.724 + \frac{3.06 S_v / S_w}{1 + \cos \Lambda_{CLM}} + \frac{0.4 Z_w}{\partial \epsilon_{max}} + 0.009 \frac{A_w}{\text{Aspect Ratio wing}}$$

$$\left(1 + \frac{\partial \sigma}{\partial \beta}\right) \eta_v = 0.724 + \frac{3.06 (221.567 / 1024)}{1 + \cos (48.7)} + \frac{0.4 (5)}{7.02} + 0.009 (8.6)$$

$$= 1.48523$$

$$(C_{n_p})_v = (0.88) (0.03963) (1.485) \left(\frac{221.6}{1023.95}\right) \left(\frac{24.36}{93.84}\right) = 0.00291 / \text{deg}$$

$$C_{L\alpha_v} = \frac{2\pi A_{eff}}{2 + \sqrt{\frac{A_v^2 \beta^2}{K^2} \left(1 + \frac{\tan^2 \Lambda_{CLM}}{\beta^2}\right) + 4}} = 0.03963 / \text{deg}$$

$$A_{v,eff} = \left(\frac{A_v(B)}{A_v}\right) A_v \left[1 + K_H \left(\frac{A_v(HB)}{A_v(B)} - 1\right)\right]$$

$$= (1.45) (1.56) \left[1 + 1.05 (0.9 - 1)\right]$$

$$= 2.262 (0.895) = 2.02449$$

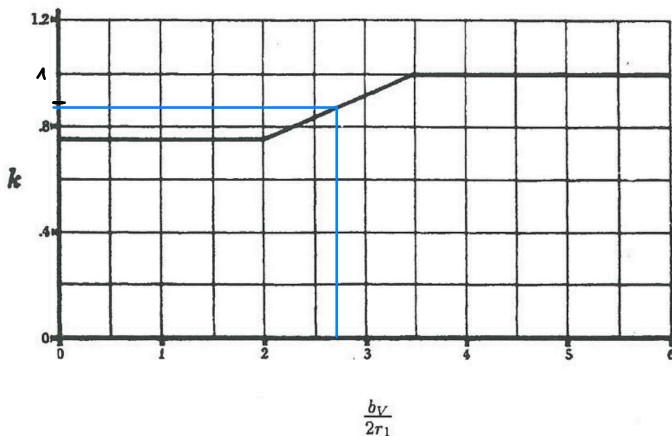


Fig. 3.75 Empirical parameter k as a function of $b_v / 2r_1$ (Ref. 1).

$$\frac{b_v}{2r_1} = \frac{20.3}{2 (3.74)} = 2.71$$

↳ Avg radius of fuselage sections under vert. tail

$$k \approx 0.88$$

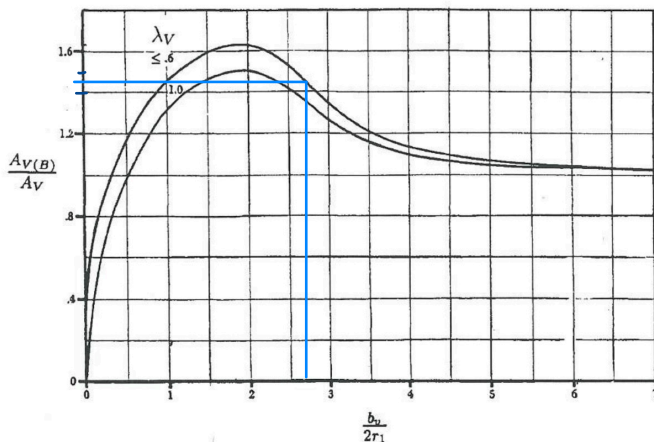


Fig. 3.77 Empirical parameter $A_{V(B)}/A_V$ as a function of $b_V/2r_1$ (Ref. 1).

$$\frac{b_V}{2r_1} = \frac{20.3}{2(3.74)} = 2.71$$

$$\lambda_V = 0.379$$

$$\frac{A_V(B)}{A_V} = 1.45$$

$$\frac{z_H}{b_V} = \frac{5}{20.3} = 0.246 \approx 0.25$$

$$\frac{x}{c_V} = \frac{2.1675}{8.67} = 0.25$$

$$\frac{A_V(HB)}{A_V(B)} \approx 0.9$$

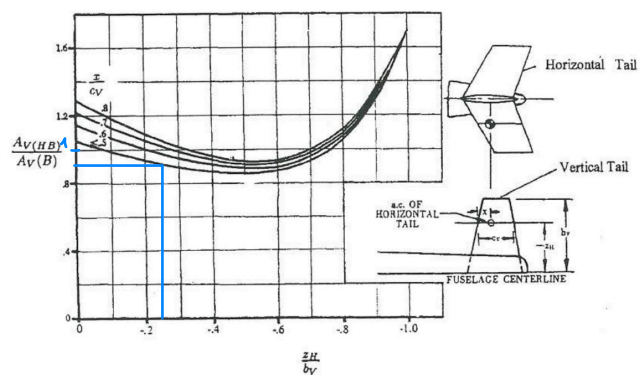


Fig. 3.78 Empirical parameter $A_{V(HB)}/A_{V(B)}$ as a function of z_H/b_V (Ref. 1).

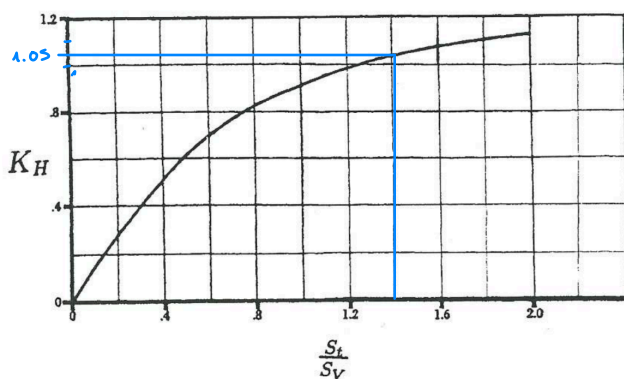


Fig. 3.79 Empirical parameter K_H as a function S_t/S_V (Ref. 1).

$$\frac{S_t}{S_V} = \frac{306.383}{221.567} = 1.38 \approx 1.4$$

$$K_H \approx 1.05$$

$$(C_{n\beta})_{TOTAL} = 0.00178 / \text{deg}$$

Lateral Stability

Fuselage Contribution

Negligible

V.T. Contribution

$$(C_{L\beta})_v = -(0.88)(0.03963)(1.485) \left(\frac{221.6}{1023.95} \right) \left(\frac{5.1 \cos 7 - 244 \sin 7}{20.3} \right)$$

$$= -0.001155862 / \text{deg}$$

Total Lateral Stability: $(C_{L\beta})_{\text{TOTAL}} = -0.00116 / \text{deg}$

Neutral Point

$$C_{m\alpha} = C_{m\alpha,f} + C_{L\alpha,w} \bar{X}_a - C_{L\alpha,t} \left(1 - \frac{d\varepsilon}{d\alpha} \right) \eta \bar{V}$$

$$\bar{X}_a = \bar{X}_{cg} - \bar{X}_{ac}$$

Conditions for Neutral Point $\begin{cases} \bar{X}_{cg} = N_0 \\ C_{m\alpha} = 0 \end{cases}$

$$0 = C_{m\alpha,f} + C_{L\alpha,w} (N_0 - \bar{X}_{ac}) - C_{L\alpha,t} \left(1 - \frac{d\varepsilon}{d\alpha} \right) \eta \bar{V}$$

$$N_0 = \frac{C_{L\alpha,t} \left(1 - \frac{d\varepsilon}{d\alpha} \right) \eta \bar{V} - C_{m\alpha,f} + C_{L\alpha,w} \bar{X}_{ac}}{C_{L\alpha,w}}$$

→ From Deliverable 1:

$$\bar{N}_0 = \frac{4.0127 \left(1 - 0.2865 \right) (1) (0.80822) - 0.05188 + 5.0464 \left(\frac{27}{11.61} \right)}{5.0464}$$

$$= 2.773$$

$$X_N = (2.773)(11.61) = 32.204 \text{ ft}$$

$$X_{cg} = 25.64 \text{ ft}$$

$$\text{Static Margin} = \frac{X_N}{\bar{c}_w} - \frac{X_{cg}}{\bar{c}_w} = \frac{32.204 - 25.64}{11.61} = 0.565$$

$$SM = 56.5 \% \bar{c}$$

Sketches

$$S_{B,5} = 551.6 \text{ ft}^2$$

